

Compiler Lab Work-7 Report

Title: English to SQL Translator using Lex and Yacc

Objective: To design a system that converts simple English sentences into SQL queries using compiler design principles.

1. Grammar Design

We designed a Context-Free Grammar (CFG) to parse simple English queries related to a Student database. The grammar supports SELECT and UPDATE queries with conditions.

2. Tools Used

Flex (Lex) for lexical analysis and Bison (Yacc) for syntax analysis.

3. Working

The lexer identifies tokens like keywords (show, give), attributes (name, roll, cpi), and operators (more than, less than).

The parser validates sentence structure and generates SQL queries using Syntax Directed Translation.

4. Example

Input: Please show name students whose cpi is more than 7

Output: SELECT name FROM Student WHERE cpi > 7;

5. Conclusion

The project demonstrates how compiler techniques can be applied to natural language processing in a limited domain.